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Lem nucleo dirichlet no es nucleo sumabilidad

Lema 1. $\forall N \in \mathbb{N} : \int_{-1/2}^{1/2} |D_N(x)| \, \mathrm{d}x \geqslant C \log(N)$ *para alguna constante* $C > 0$.

***Demostración*:** Por el `lem-formula-nucleo-dirichlet/`Lema 1, sabemos que D_N es una función par y sus ceros en $\left[0, 1/2\right)$ son los $x \in \mathbb{R}$ tales que $(2N+1)\pi x = k\pi \iff x = \frac{k}{2N+1}$ para $k = \mathbb{N}_N$. Por tanto,

$$\begin{aligned}
\int_{-\frac{1}{2}}^{\frac{1}{2}} |D_N(x)| \, \mathrm{d}x &\overset{\substack{\text{par}\\ \downarrow}}{=} 2 \int_0^{\frac{1}{2}} |D_N(x)| \, \mathrm{d}x = 2 \sum_{k=0}^{N-1} \int_{\frac{k}{2N+1}}^{\frac{k+1}{2N+1}} |D_N(x)| \, \mathrm{d}x \\
&\geqslant 2 \sum_{k=0}^{N-1} \int_{\frac{k}{2N+1}}^{\frac{k+1}{2N+1}} \frac{|\sin((2N+1)\pi x)|}{\sin\left(\pi \frac{k+1}{2N+1}\right)} \, \mathrm{d}x \quad \left(\sin(\pi x) \text{ es creciente en } \left[0, 1/2\right)\right) \\
&= 2 \sum_{k=0}^{N-1} \frac{1}{\sin\left(\pi \frac{k+1}{2N+1}\right)} \underbrace{\int_k^{k+1} |\sin(\pi y)| \frac{\mathrm{d}y}{2N+1}}_{=\frac{2}{\pi}} \quad (y = (2N+1)x) \\
&= \frac{2}{2N+1} \sum_{k=0}^{N-1} \frac{1}{\sin\left(\pi \frac{k+1}{2N+1}\right)} \frac{2}{\pi} \geqslant \frac{4}{2N+1} \frac{1}{\pi} \sum_{k=0}^{N-1} \frac{1}{\frac{k+1}{2N+1}} = \frac{4}{\pi} \sum_{k=0}^{N-1} \frac{1}{k+1} \\
&\geqslant \frac{4}{\pi} \int_1^N \frac{1}{x} \, \mathrm{d}x = \frac{4}{\pi} \log(N).
\end{aligned}$$

Por tanto, podemos tomar $C = \frac{4}{\pi}$ y la desigualdad queda demostrada. ■

Observación 1. El **??** muestra que el núcleo de Dirichlet no es un núcleo de sumabilidad según la `nucleo-sumabilidad/??`, ya que no satisface la condición (2).

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